Example: QP Formulation

Objective Function:

Minimize 0.5 x^2 + 3x + 4y

Standard QP form: ½ [x^T P x] + q^T x

Subject to: Gx ≤ h

x ≥ 0 => -x ≤ 0

y ≥ 0 => -y ≤ 0

x + 3y ≥ 15 => -x - 3y ≤ -15

2x + 5y ≤ 100

3x + 4y ≤ 80

G =

```
[-1  0 ]
[ 0 -1 ]
[-1 -3 ]
[ 2  5 ]
[ 3  4 ]
```

h =

```
[  0 ]
[  0 ]
[-15 ]
[100 ]
[ 80 ]
```